

HIGH-PRESSURE GATHERING LINE - SEAMLESS PIPE QUALITY PLAN

Pipe Description: 12.75" (323.9 mm) x 12.70 mm, Grade X52Q PSL 2, SMLS (Seamless Pipe) | **Governing Standard:** API Specification 5L 47th Edition / ISO 3183 PSL 2

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.16%, P <= 0.020%, S <= 0.010%, CE_IIW <= 0.43	API 5L Tablo 5	H/W
2	Product Chemical Analysis	Two analyses per test unit (lot <= 100 pipes)	Pipe body sample	ISO 14284	C <= 0.16%, P <= 0.020%, S <= 0.010%, CE_IIW <= 0.43	API 5L Tablo 5	W
3	Pipe Body Tensile Test	Once per test unit after Q&T; heat treatment	Transverse body coupon	ASTM A370	Rt0.5: 360-530 MPa, Rm: 460-760 MPa, Af >= 23%, Y/T <= 0.93	API 5L Tablo 7	H/W
4	CVN Charpy Impact Test (Body) at 0 °C	1 set (3 specimens) per test unit at 0 °C	Transverse body Charpy V	ASTM A370	At 0 °C: Min Average 40 Joules, Min Individual 30 Joules	API 5L Tablo 8	H/W
5	Mill Hydrostatic Pressure Test	Each pipe (100% all seamless pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 240 bar, Duration: MINIMUM 5 SECONDS (D < 457 mm)	API 5L 10.2.6.2	H/W
6	Seamless Pipe Body Full NDT (Flux Leakage / UT)	100% full length and 360° circumference	Full pipe body	ISO 10893-1 / 10893-3	100% automated Ultrasonic or Magnetic Flux Leakage for longitudinal/transverse flaws	API 5L Ek E	H/W
7	Pipe Ends UT Laminar Testing	100% of pipe ends (50 mm band)	Both pipe ends	API 5L Annex E.8	50 mm circumferential band on both ends; no laminar defect > 6.0 mm	API 5L Ek E.8	W
8	Dimensional & Wall Thickness Verification	100% of all pipes (Ultrasonic wall check)	Pipe body & ends	API 5L 10.2.8	Diameter: +-0.75%; Wall thickness: -12.5% / +15.0% (SMLS Table 11)	API 5L Tablo 10/11	W
9	Visual Surface & Bevel Geometry	100% internal and external surface	Full pipe length	API 5L 10.2.7	No crack, no gouge, no defect > 12.5% t; Bevel: 30° (+5°/-0°), root face 1.6 mm	API 5L 9.12 & 9.14	W